

Characterization of SOTA CKGE methods

ISWC'26

CKGE Techniques:

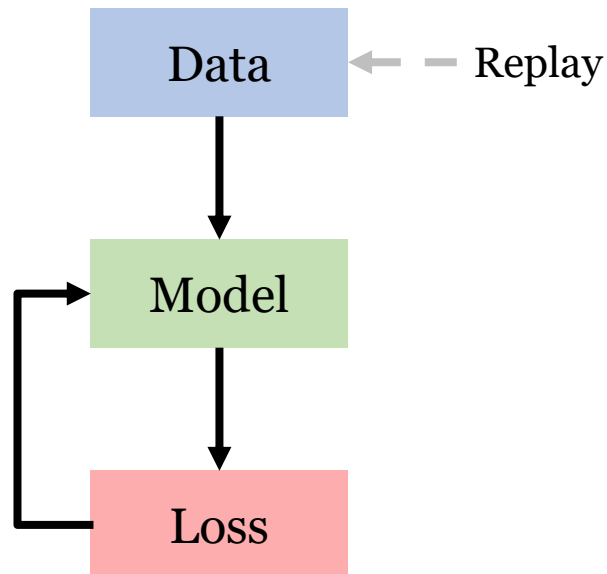
The goal of CKGE is to update KGE of evolving KGs by **acquiring new knowledge** without **losing the previously acquired**. This is done by proposing techniques that are applied during the training cycle, which are typically seen as **additional terms in the loss function** when compared to fine-tuning.

$$Loss = Loss_{new} + Extra$$

KGE model (e.g., TransE)
loss over the new triples.

CKGE Techniques:

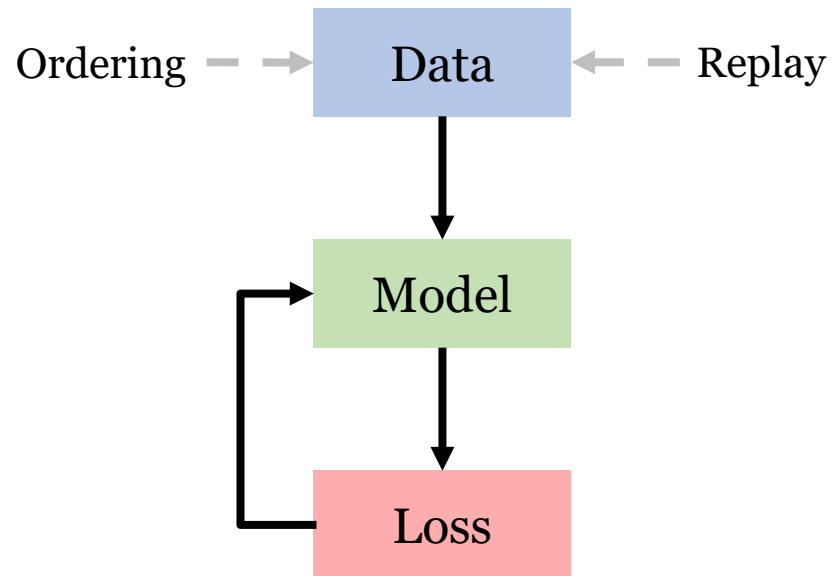
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Replay strategies focus on allocating a memory buffer to **store triples** from past snapshots (i.e., episodic memory), which are then **combined** with new triples during training. **In some CL context, reusing past data is not allowed.**

CKGE Techniques:

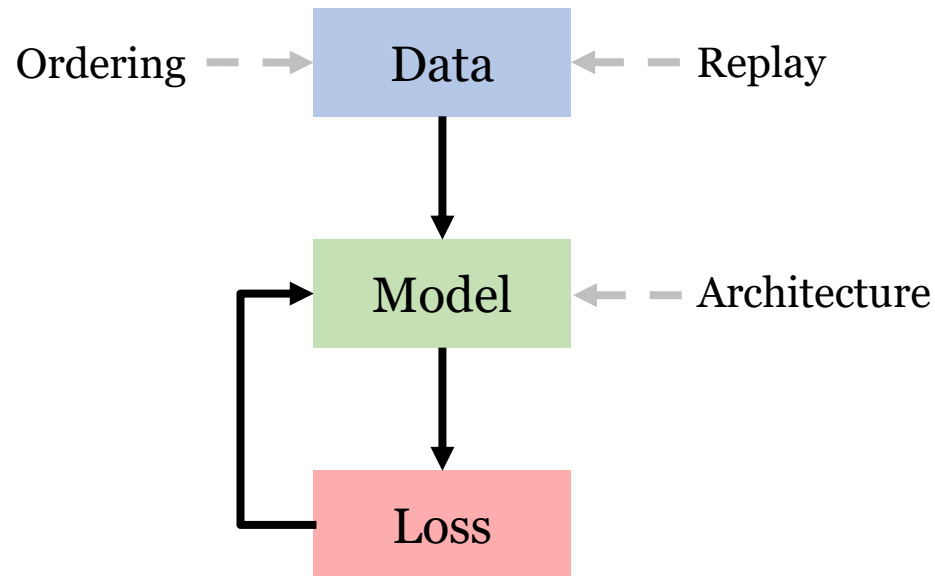
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Ordering strategies consist on defining the order in which **data** will be fed to the model.

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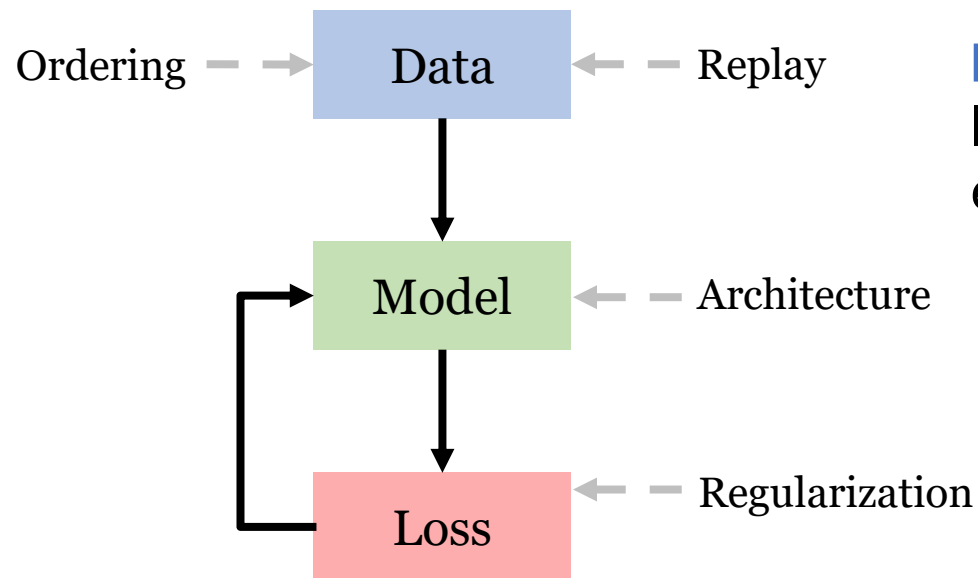
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Architecture strategies modify the **KGE model** to accommodate new information.

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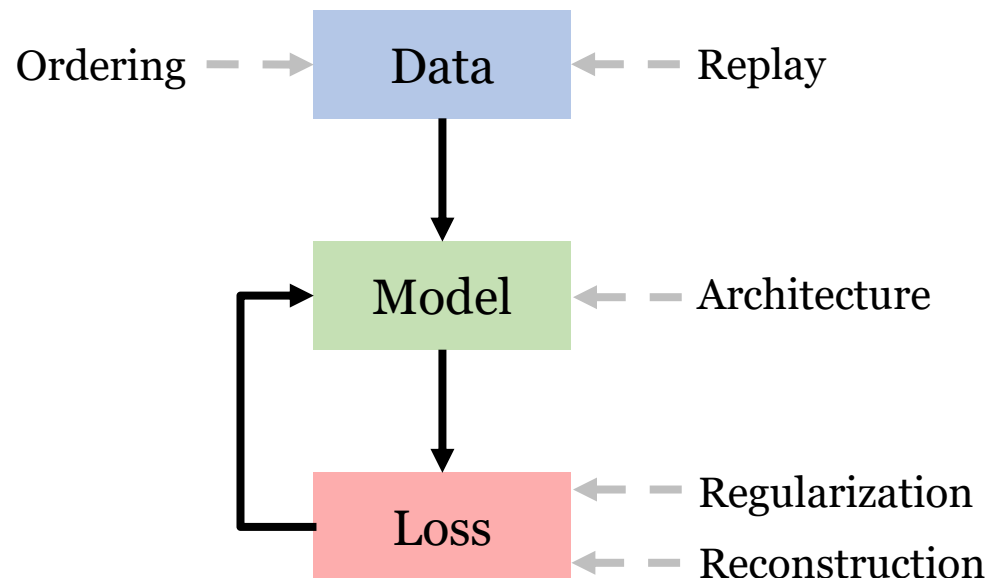
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Regularization strategies design terms for the loss function that **penalize changes** of existing parameters.

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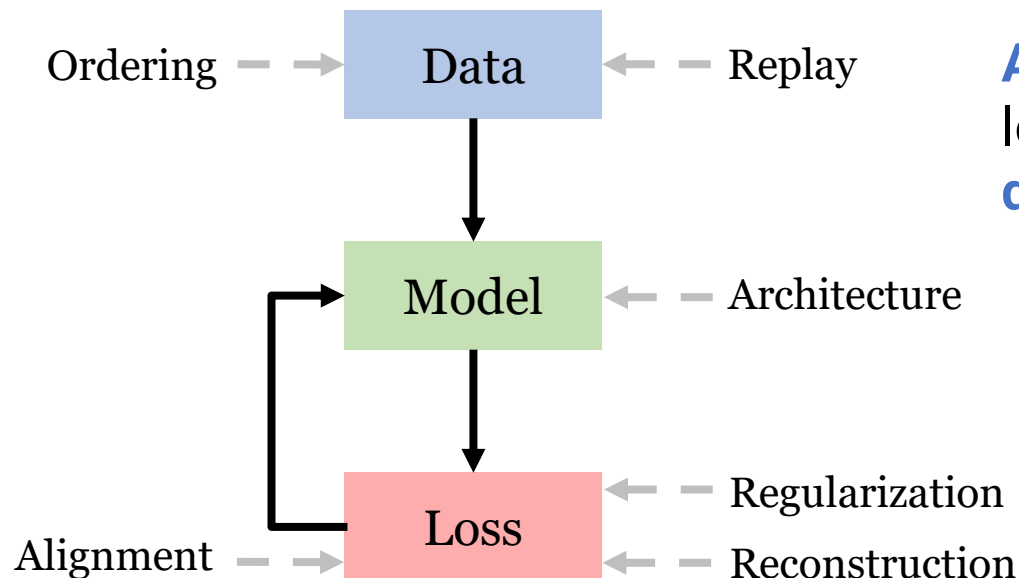
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Reconstruction strategies define extra terms for the loss function that try to **derive** existing embeddings from additional (e.g., neighbors) information.

CKGE Techniques:

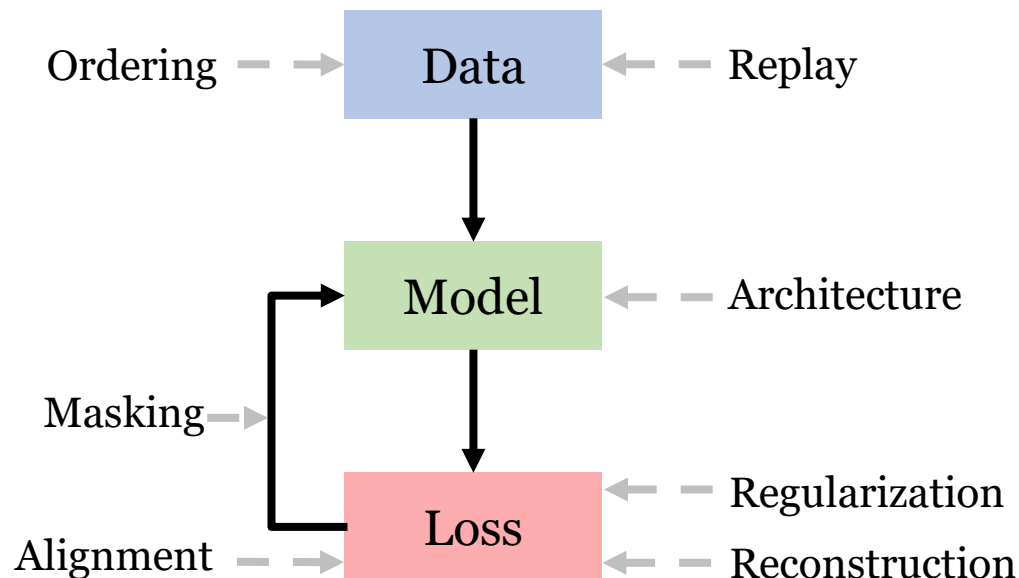
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Alignment strategies design terms for the loss function that **penalize changes** in the **direction** of the embeddings.

CKGE Techniques:

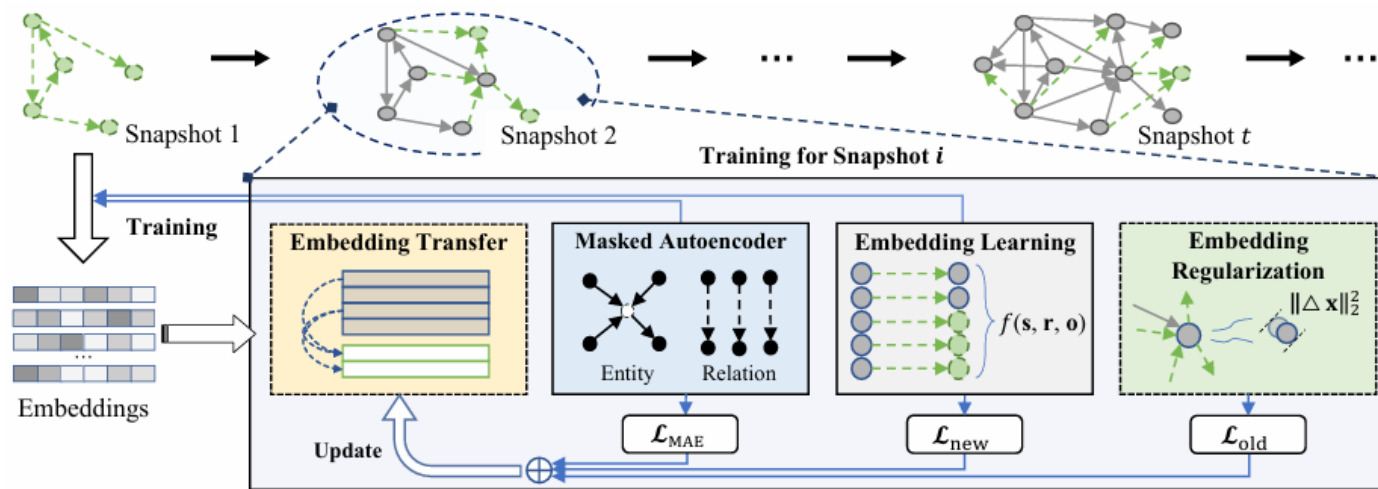
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Masking strategies define which parameters (and even which **dimensions** of them) are allowed to change.

CKGE Approaches: LKGE

LKGE was the **first KG-specific** method proposed. It's main building blocks are a **Masked Autoencoder** and **Embedding Regularization**.



[1] Y. Cui, Y. Wang, Z. Sun, W. Liu, Y. Jiang, K. Han, W. Hu, Lifelong embedding learning and transfer for growing knowledge graphs, in: AAAI, 2023.

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In the **Masked Autoencoder**, the embedding value is **reconstructed** with the expected position by the KGE model based on the triples each entity/relation participates in.

$$\mathcal{L}_{\text{MAE}} = \sum_{e \in \mathcal{E}_i} \|\mathbf{e}_i - \bar{\mathbf{e}}_i\|_2^2 + \sum_{r \in \mathcal{R}_i} \|\mathbf{r}_i - \bar{\mathbf{r}}_i\|_2^2.$$

$$\bar{\mathbf{e}}_i \approx \frac{\sum_{j=1}^{i-1} |\mathcal{N}_j(e)| \mathbf{e}_{i-1} + \sum_{(s,r,o) \in \mathcal{N}_i(e)} f_{\text{sub}}(\mathbf{r}_i, \mathbf{o}_i)}{\sum_{j=1}^{i-1} |\mathcal{N}_j(e)| + |\mathcal{N}_i(e)|}$$

TransE

$$t \approx h + r$$

$$h \approx t - r$$

CKGE Approaches: LKGE

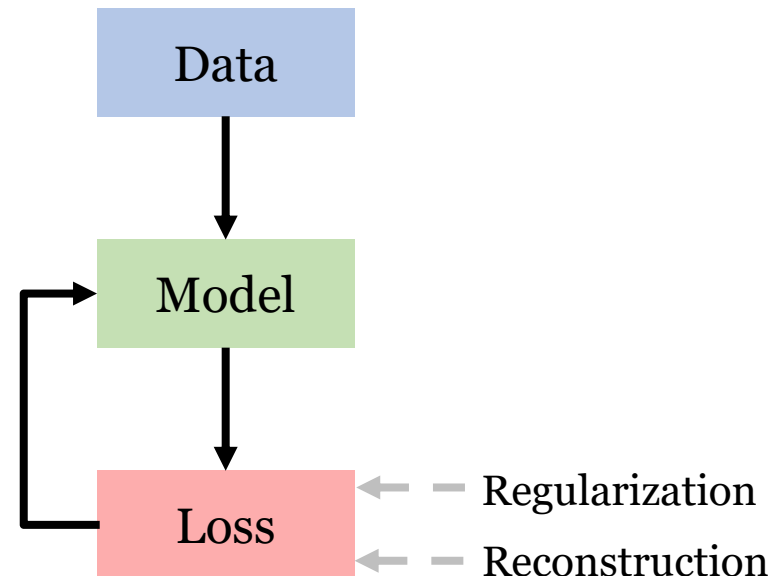
LKGE was the **first KG-specific** method proposed. It's main building blocks are a **Masked Autoencoder** and **Embedding Regularization**.

In **Embedding Regularization**, the changes in the embeddings are penalized via a **weighted L2 regularization**. The weight is **inversely proportional** to the total of times the entity appeared in the snapshot compared to the previous ones.

$$\mathcal{L}_{\text{old}} = \sum_{e \in \mathcal{E}_{i-1}} \omega(e) \|\mathbf{e}_i - \mathbf{e}_{i-1}\|_2^2 + \sum_{r \in \mathcal{R}_{i-1}} \omega(r) \|\mathbf{r}_i - \mathbf{r}_{i-1}\|_2^2$$

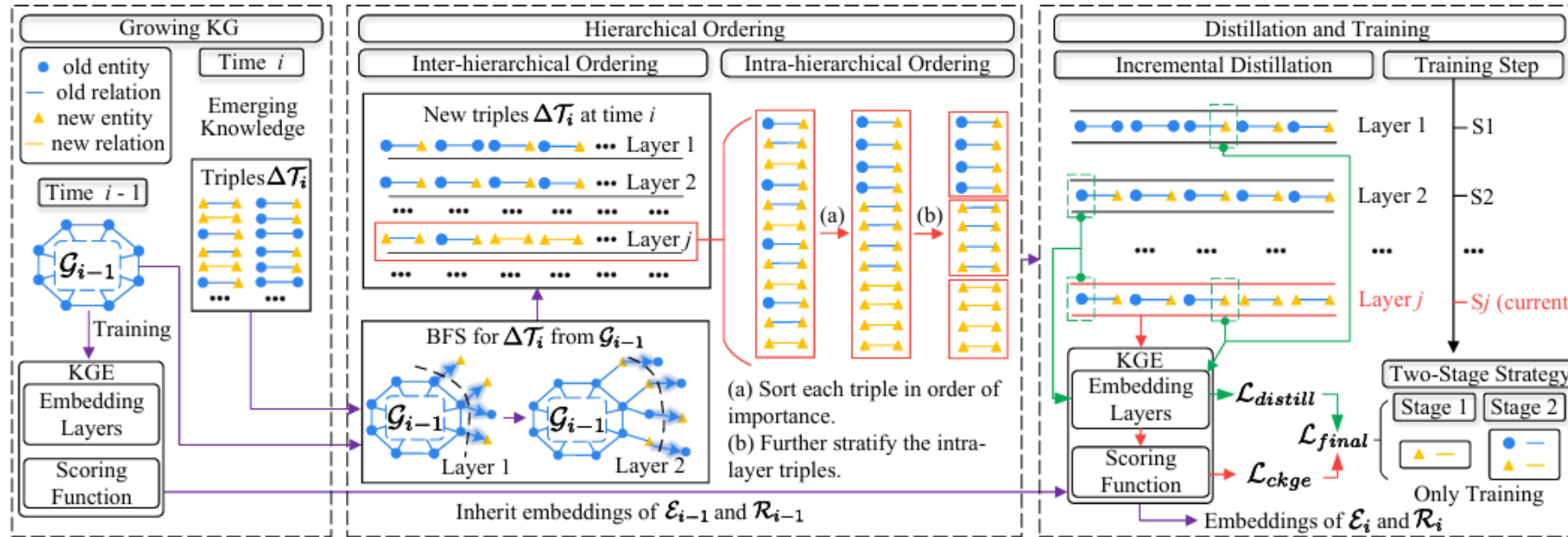
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CKGE Approaches: IncDE

In IncDE, the main innovations are in **triple ordering** and **incremental distillation**.



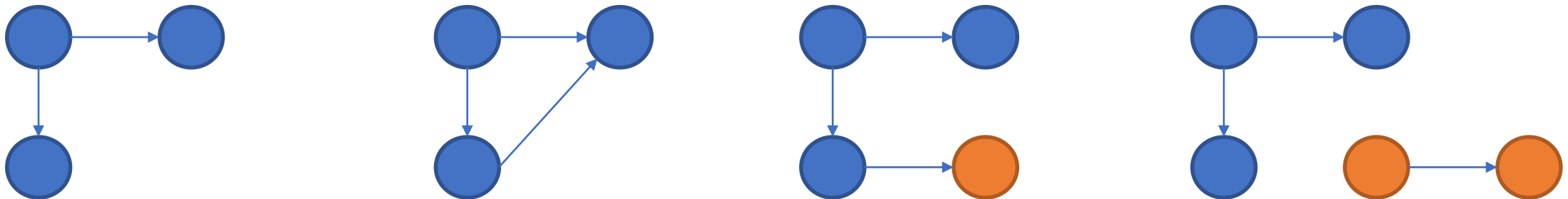
[2] J. Liu, W. Ke, P. Wang, Z. Shang, J. Gao, G. Li, K. Ji, Y. Liu, Towards continual knowledge graph embedding via incremental distillation, in: *Proceedings of the AAAI Conference on Artificial Intelligence*,

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For **triple ordering**, there are two ordering steps:

- First triples are ordered based on the **distance** their **entities** and **relation** and the **graph**.



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For **triple ordering**, there are two ordering steps:

- First triples are ordered in layers based on the **distance** their **entities** and **relation** and the **graph**.
- Among a layer, triples are ordered based on the **importance** of the their entities.

$$f_{nc}(e) = \frac{f_{neighbor}(e)}{N - 1}$$

$$f_{bc}(r) = \sum_{s,t \in \mathcal{E}_i, s \neq t} \frac{\sigma(s, t|r)}{\sigma(s, t)}$$

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The training stage is done in **batches (e.g., layers)** according to this ordering.

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Incremental distillation is a regularization technique that compares the learned representation with the one obtained in the **nearest layer**.

$$\mathcal{L}_{distill}^k = \begin{cases} \frac{1}{2}(\mathbf{e}'_k - \mathbf{e}_k)^2, & |\mathbf{e}'_k - \mathbf{e}_k| \leq 1 \\ |\mathbf{e}'_k - \mathbf{e}_k| - \frac{1}{2}, & |\mathbf{e}'_k - \mathbf{e}_k| > 1 \end{cases}$$

The regularization loss for each entity is weighted by **centrality metrics**.

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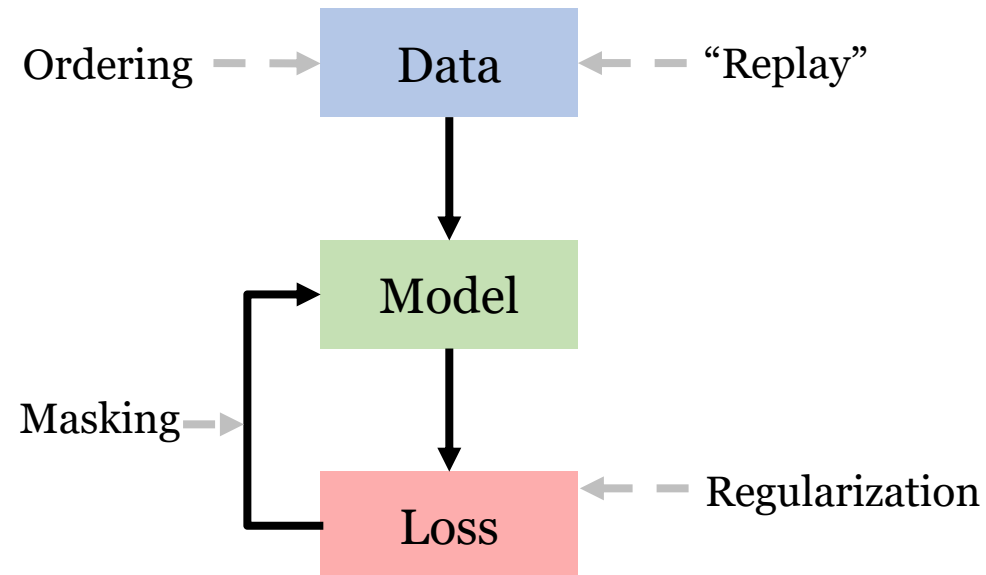
In IncDE, the main innovations are in **triple ordering** and **incremental distillation**.

Additionally, they introduce a **two-stage** training process:

- First, to better preserve information, **old** entity and relation **embeddings are frozen**, and only the new ones are allowed to change.
- Then, the same **training cycle is repeated**, allowing all embeddings to change.

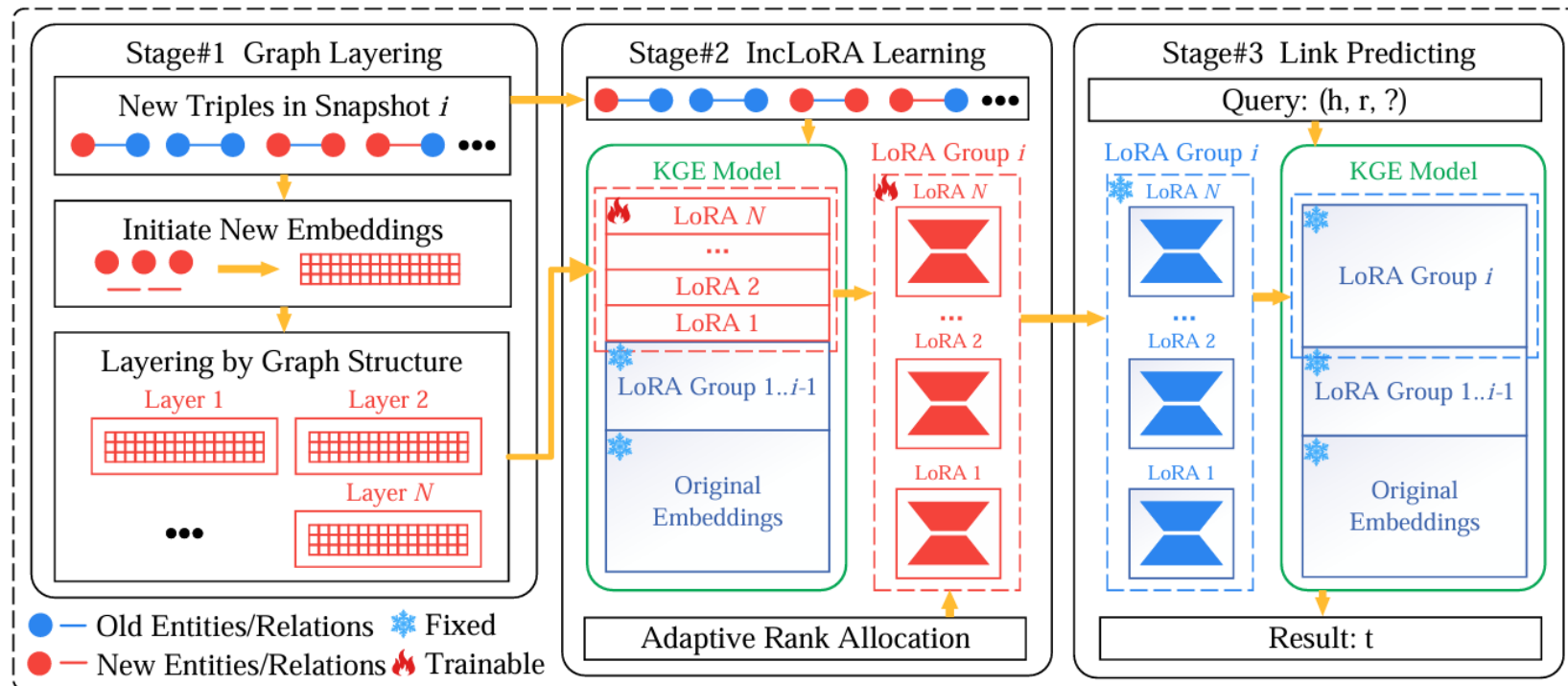
CKGE Approaches: IncDE

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CKGE Approaches: FastKGE

This method's goal is to **reduce training time** while maintaining **competitive performance**.



[3] Liu, J., Ke, W., Wang, P., Wang, J., Gao, J., Shang, Z., Li, G., Xu, Z., Ji, K., Li, Y., 2024b. Fast and continual knowledge graph embedding via incremental lora, in: IJCAI.

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This method's goal is to **reduce training time** while maintaining **competitive performance**. It does so by learning **Low Rank Adapters** for different groups of new entities:

1. First, new entities are partitioned into different layers (similar to IncDE) based on the proximity from the graph and entity centrality metrics.

$$\overline{E} = \{E_1, E_2, \dots, E_N\}$$

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2. Then, an Incremental Low Rank Adapter is learned for each layer. $\mathbf{E}_k = \mathbf{A}_k \mathbf{B}_k$

The rank of each layer is assigned based on the importance (degree centrality) of all the entities in that group.

$$Sum_k = \sum_{e \in E_k} f_{dc}(e) \quad Avg_{dc} = \frac{\sum_{k \in N} Sum_k}{N} \quad r_k = \frac{r_{base} \cdot Sum_k}{Avg_{dc}}$$

CKGE Approaches: FastKGE

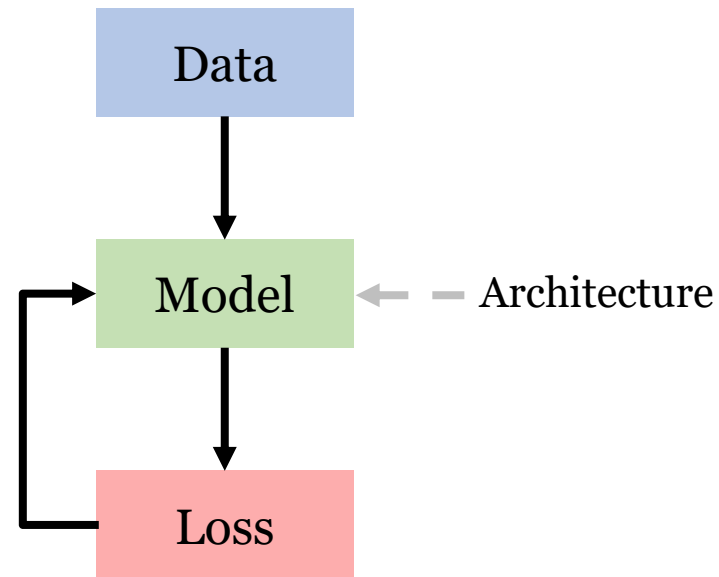
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The rank of each layer is assigned based on the importance (degree centrality) of all the entities in that group
3. Finally, the new embeddings are trained while the old ones remain frozen. !

$$\mathbf{E}_{all} = \text{concat}(\mathbf{E}_{origin}, \{\mathbf{G}_j^{\mathbf{E}} | 1 \leq j \leq i\})$$

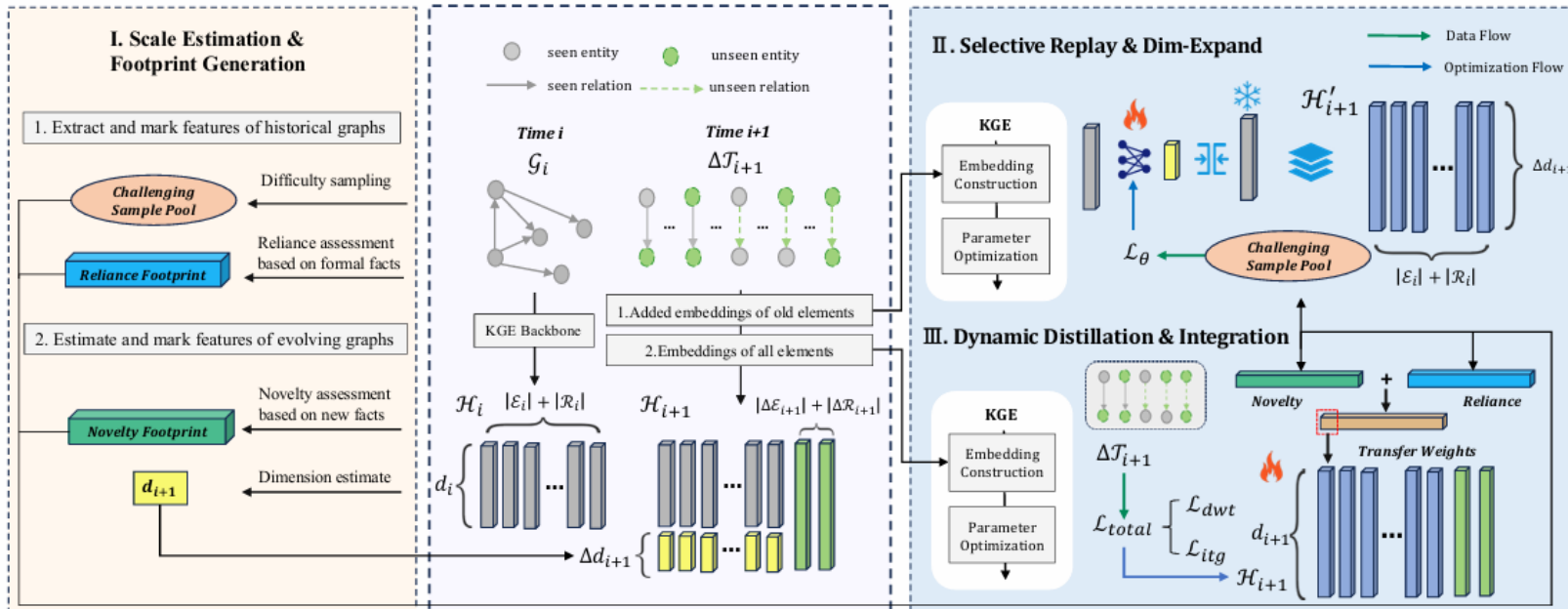
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CKGE Approaches: SAGE

This method argues that the **embedding dimension** should be **expanded** to accommodate **new knowledge**.



[4] Li, Y., Zhang, L., Yan, H., Zhao, T., Ma, Z., Huang, M., Liu, J., 2025. Sage: Scale-aware gradual evolution for continual knowledge graph embedding, in: *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*

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For that, first the **new dimension** is computed:

$$d_{i+1} = \begin{cases} r \cdot d_i, & \text{if } r \cdot d_i \leq y_{\min}, \\ y_{\min}, & \text{if } \frac{y_{\min}}{r} < d_i \leq y_{\min}, \\ y, & \text{if } y_{\min} < d_i \leq y, \\ d_i + \text{step}, & \text{if } y < d_i \leq y_{\max}, \\ d_i, & \text{if } y_{\max} < d_i, \\ y_{\max}, & \text{otherwise.} \end{cases}$$

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Then, old embeddings need to be **expanded** to the new dimension. This is done by a learnable transformation that obtains the missing dimensions. This transformation is trained with a small sample of **previously challenging triples** (based on model's uncertainty).

$$\mathcal{H}'_{i+1} = \bigcup_{j=1}^{|\mathcal{E}_i|+|\mathcal{R}_i|} \left\{ \text{concat}(\mathbf{h}_i^j, f_{\theta}(\mathbf{h}_i^j)) \right\}$$

CKGE Approaches: SAGE

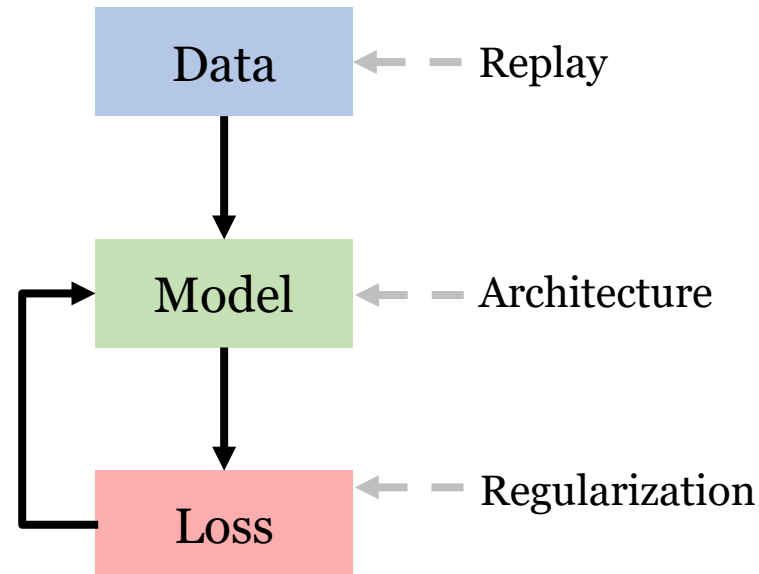
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Once expanded, training proceeds as usual, incorporating also a **regularization term** in the loss function, which is weighed by novelty (in how many new triples the entity appears) and reliability (in how many old triples the entity appears, and which was the MRR score from them):

$$\mathcal{L}_{dwt} = \sum_{x_j \in (\mathcal{E}_i \cup \mathcal{R}_i)} \frac{f_r(x_j)}{f_n(x_j)} \|\mathbf{h}_j - \mathbf{h}'_j\|_2^2$$

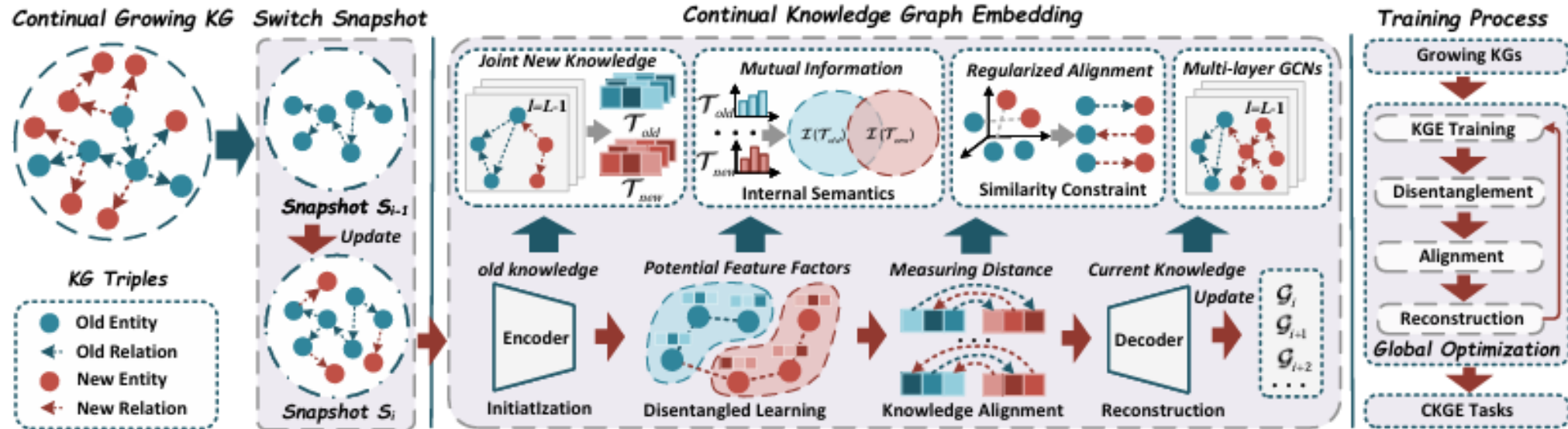
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CKGE Approaches: DebiasedKGE

The main new contribution are in **knowledge misalignment**, caused by new knowledge altering the relational structure of existing knowledge, and **disentanglement**.



[5] Zhu, J., Fu, B., Duan, G., 2025a. Debiasedkge: Towards mitigating spurious forgetting in continual knowledge graph embedding, in: Proceedings of the 34th ACM International Conference on Information and Knowledge Management

CKGE Approaches: DebiasedKGE

The main new contribution are in **knowledge misalignment**, caused by new knowledge altering the relational structure of existing knowledge, and **disentanglement**.

For alignment, a new penalization is introduced to the changes in the embeddings that affect the direction (i.e., **directional shifts**).

$$\mathcal{L}_{\cos} = \frac{1}{N} \sum_{m=1}^N \left(1 - \cos(e_m^{\text{new}}, e_m^{\text{old}}) \right)$$

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For **disentanglement**, K different subspaces are generated, encouraging each one to be independent in order to enhance **semantic separability**.

$$\mathcal{I}(\mathcal{T}_i) = \sum_{k=1}^K \sum_{k'=k+1}^K \left(\mathbb{D}_{\text{KL}} \left(\tilde{\mathbf{E}}_i^{(k)} \parallel \tilde{\mathbf{E}}_i^{(k')} \right) + \mathbb{D}_{\text{KL}} \left(\tilde{\mathbf{R}}_i^{(k)} \parallel \tilde{\mathbf{R}}_i^{(k')} \right) \right)$$

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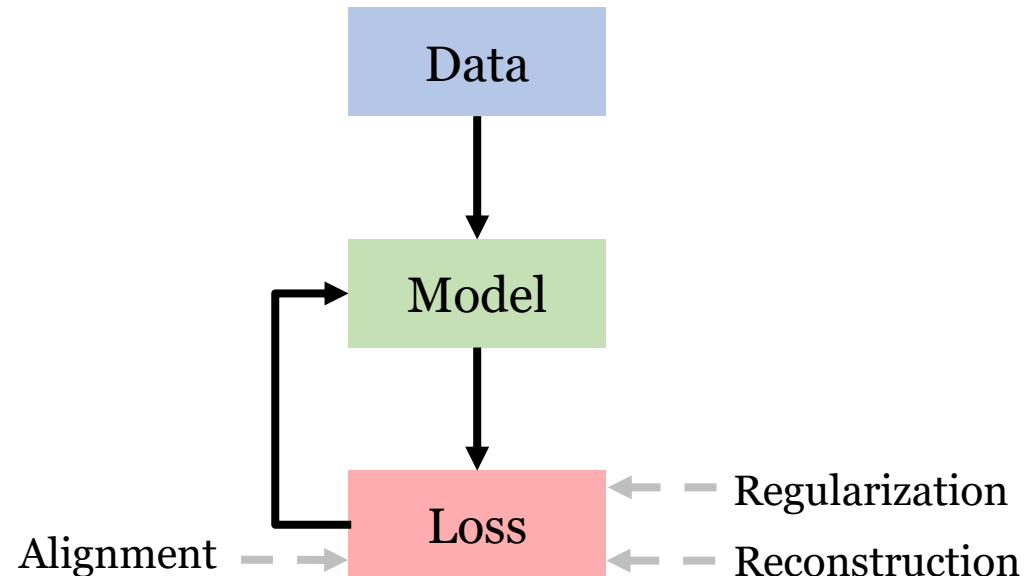
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Additionally, DebiasedKGE also has a **knowledge reconstruction** module based on Graph Neural Networks.

$$\mathbf{E}_i^{(l+1)} = \sigma \left(\text{GCN}_e^{(l)} \left(\mathbf{E}_i^{(l)}, \mathcal{A}_i, \mathcal{R}_i \right) \right) + \mathbf{E}_i^{(l)}$$

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Questions?